

Data Programming in Python | BAIS:6040

Homework 2. Functions and Branching

In this homework, you will be constructing a monthly budget for post-graduation using user-defined functions, input, and if-else statements.

Instructions

- Fill in the Python statements needed to complete all the questions below.
- Program everything inside this notebook.
- Use **only one cell per question**.
- Each code cell must be run with no errors.
- Do not contain any unnecessary code in your answer. (Only the final outcome will be regarded as your answer.)
- If the question requests that you store information in a certain variable, use the specific variables name (case sensitive).
- All problems must be solved **in a programmable way**.
- When you are done, make sure to save your notebook and then click Restart & Run All before submission to create a clean version.

Questions (40 points total)

1. (3 pts) You will be using the `input` function to get values of variables from the user. `x = input("Prompt they will see")`. The `input` function by default returns a string. Practice getting input for a str, int, and float with the variables below. Remember, you will need to convert your input to a number data type for the last two.

```
In [ ]: # Your answer here
int_num = int(input("Type an integer:"))
str_value = input("Type a string:")
float_num = float(input("Type a float:"))
```

```
In [ ]: # Check your answer here. (Do not make any change to this cell. Just run this cell.
print(type(int_num))
print(type(str_value))
print(type(float_num))
```

2. (2 pts) Use input to get a float number take-home pay, which is your projected monthly take-home pay after taxes (net pay in \$: 5000 for example). This number can of course be made up and will be entered in by the user like in Question 1.

```
In [ ]: #Your answer here.
pay = float(input("What is your projected take-home pay? $"))
```

https://www.w3schools.com/python/python_string_formatting.asp

<https://www.investopedia.com/terms/f/futurevalue.asp>

3. (8 pts) Define a function `roth_ira` that takes one argument `budget`, which is your current balance before investing in your monthly Roth IRA retirement. Within the function, use `input` to ask for `contribution` from the user. The most you can contribute is 7000 per year which equals \$583.33 per month. Your `contribution` also can't be negative. Set up if/elif/else statements that returns the balance and warning strings if `contribution` is > 583.33 or < 0. Otherwise, it returns the money you have left, `balance - contribution` and the f-string of the future value of your investment for 1 month 30 years from now! Assume a 10% annual rate of return and use the future value formula.

https://www.w3schools.com/python/python_string_formatting.asp

<https://www.investopedia.com/terms/f/futurevalue.asp>

```
In [ ]: #In this cell define your roth_ira function
def roth_ira (balance):
    contribution = float(input("How much will you contribute to your Roth IRA? $"))
    fv = contribution * (1+.1)**30
    if contribution < 0:
        return balance, "You can't contribute a negative value."
    elif contribution > 583.33:
        return balance, "You can't contribute more than $583.33"
    else:
        return balance - contribution, f"You will have ${fv:.2f} in 30 years if you
```

```
In [ ]: #Running this cell and giving 500 as input should give you the following output
money_left, future_money = roth_ira(pay)
print(money_left)
print(future_money)
```

```
"""
How much will you contribute to your Roth IRA? $500
500.0
You will have $8724.70 in 30 years if you invest $500.0 now.
"""
```

```
#Running this cell and giving -10 as input should give you the following output
"""
How much will you contribute to your Roth IRA? $-10
1000
```

```
You can't contribute a negative value.
"""
```

4. (3 pts) Practice working with default arguments in your function definition. In the `practice_fun` function below, set both `y` and `z` to have default values of 3 and 4 respectively. `x` is the first argument and will not have a default value. Inside of the function perform the operation `answer = x + y * z` and return that answer. See the cell with the function call below to see expected output depending on how many arguments you enter.

```
In [ ]: def practice_fun(x, y=3, z=4):
        answer = x + y * z
        return answer
```

```
In [ ]: #You just need to run this cell, you do not need to change its contents.
ans1 = practice_fun(2)
print(ans1)
#Expected output 14
ans2 = practice_fun(2,4)
print(ans2)
#Expected output 18
ans3 = practice_fun(2,4,6)
print(ans3)
#Expected output 26
```

5. (8 pts) You will be creating a `loan_calculator` function that calculates the student loan monthly payment shown at this website. We will assume a standard repayment plan of 10 years and 120 payments.

<https://lendedu.com/blog/student-loan-payment-calculator/>

Outside of the function, get the annual rate (.07 for 7 percent) and principal. When defining the function, you have 3 arguments. The first argument is your `balance`, which is the same as Problem 3 or what is left in your monthly budget. The other 2 arguments are `rate` and `loan_amt`, which we got as input from the user in the cell before. The default arguments for those will be .075 and 20000.

Inside of the function, calculate the monthly loan payment based on the calculation from the website above. Return `monthly_payment` and `money_left` on your budget after you pay your student loan monthly payment.

```
In [ ]: annual_rate = float(input("Please input your interest rate as a decimal (ex. .1 for
principal = float(input("How much did you originally take out for your loan? "))
```

$$M = P \frac{r(1+r)^n}{(1+r)^n - 1} \cdot 30.5$$

```
In [ ]: def loan_calculator(balance, rate=0.075, loan_amt=20000):
        interest = (1 + rate/365.25)**(365.25*10)
        monthly_payment = loan_amt*(rate/365.25*interest)/((interest-1)*30.5
        money_left = balance - monthly_payment
        return monthly_payment, money_left
```

```
In [ ]: pmt, money_left = loan_calculator(money_left, annual_rate, principal)
```

```
In [ ]: print(pmt)
        print(money_left)
```

6. (8 pts) It is difficult to predict monthly car payments, but you may be in the market for a used or new car when graduating and starting at your new job. To simplify, create a function that takes two arguments `balance` and `car_cost`. There are many variable factors based on credit score, available financing, length of loan, but we can do a rough estimate using if-elif-else statements that follow the table below. Depending on the `car_cost` variable, return the `car_pmt` which is the monthly estimated car payment and `money_left` which is how much is left on the balance after subtracting the car payment.

Under 15000 - \$250

15000-19999 - \$300

20000-24999 - \$400

25000-29999 - \$500

30000-34999 - \$600

35000-39999 - \$700

40000 and over - \$800

```
In [ ]: cost = int(input("What is the cost of the car? $"))
```

```
In [ ]: def car_pay(balance, car_cost):
    if car_cost < 15000:
        car_pmt = 250
    elif car_cost <= 19999:
        car_pmt = 300
    elif car_cost <= 24999:
        car_pmt = 400
    elif car_cost <= 29999:
        car_pmt = 500
    elif car_cost <= 34999:
        car_pmt = 600
    elif car_cost <= 39999:
        car_pmt = 700
    else:
        car_pmt = 800

    balance -= car_pmt
    return car_pmt, balance
```

```
In [ ]: car_pmt, money_left = car_pay(money_left, cost)
print(car_pmt)
print(money_left)
```

7. (8 pts) Like car payments, it is difficult to know how much your monthly housing costs will be, but they on average take up one-third of your budget. For this function, `housing` there will be 1 argument `balance` and 3 inputs the user gives: `home_cost`, `rate`, `years`. Use an if-else statement and compound condition to return "You should rent" or "You should buy". If the rate is greater than > 5% (.05) OR you plan on staying at that place for less than < 5 years then rent, otherwise buy. In addition, subtract the `home_cost` from balance and return that as `money_left`.

```
In [ ]: def housing(balance):
    home_cost = float(input("Home cost: $"))
    rate = float(input("Interest rate (as decimal): "))
    years = int(input("Years you plan to stay in the residence: "))
    money_left = balance - home_cost
    if rate > .05 or years < 5:
        str1 = "You should rent"
    else:
        str1 = "You should buy"
    return str1, money_left
```

```
In [ ]: rent_or_buy, money_left = housing(money_left)
print(rent_or_buy)
print(money_left)
```

8. (4 pts) Americans spend on average 12.8% of their budget on food. They also spend 55.7% of their food budget on food away from home (FAFH) like dining out or takeout or food delivery. <https://www.escoffier.edu/blog/world-food-drink/consumer-dining-trend-statistics/>

In this question create a function that calculates how many FAFH meals you could buy if your food spending habits are exactly like the average American above, use the `pay` variable from Question 2 as an argument and `balance` as your 2nd argument of the function.

Inside the function, use input to get the `avg_cost` of a meal when the user eats out (\$20 for example). You will calculate how much you would spend total on FAFH based on the percentages from the start of this question multiplied by `pay`. Then divide `pay` by `avg_cost` to get the `num_meals`. Finally, calculate the money you have left after your TOTAL food budget (that 12.8% value).

Return the `num_meals` and `money_left`.

```
In [ ]: def food(pay, balance):
        avg_cost = float(input("What is the average cost of a meal when you eat out? $"))
        money_left = balance - (pay*.128)
        pay = pay * .128 * .557
        num_meals = pay/avg_cost
        return num_meals, money_left
```

```
In [ ]: num_meals, money_left = food(pay, money_left)
        print(num_meals)
        print(money_left)
```

9. (2 pts) After running all of the budget functions above, examine your `money_left` variable. Use if-else statements to print out how much money you have left or if you are over budget depending on the `money_left` value.

You do not need to do this in a user-defined function and can write your code in the cell below.

Expected output:

You are \$ ___ over budget. or

You have \$ ___ money left to spend on other items!

```
In [ ]: #Answer below
        if money_left >= 0:
            print(f"You have ${money_left} money left to spend")
        else:
            print(f"You are ${0- money_left} over budget")
```

```
In [ ]:
```